

11th grade
Term 3 — Lesson 1 Practice Activity (C5) with Solutions

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Evaluated criteria

- C5: Calculates hypothesis tests on the mean of a population.

Problems

P1 1. Average export clearance time

A customs agency claims the mean clearance time is $\mu_0 = 18$ hours. A random sample of $n = 64$ shipments has $\bar{x} = 19,1$ hours. Assume $\sigma = 4$ hours. At $\alpha = 0,05$, test whether the mean is greater than 18 hours.

C5

- **Hypotheses:** $H_0: \mu = 18$, $H_1: \mu > 18$.
- **Test statistic:**

$$z = \frac{19,1 - 18}{4/\sqrt{64}} = \frac{1,1}{0,5} = 2,20.$$

- **p-value:** $P(Z \geq 2,20) \approx 0,0139$.
- **Decision:** Reject H_0 since $0,0139 < 0,05$.
- **Conclusion:** There is sufficient evidence that the mean clearance time exceeds 18 hours.

P2 2. Mean monthly inflation expectation

Economists test whether the mean household inflation expectation is 6.0 %. A sample of $n = 100$ households gives $\bar{x} = 5,7$ %. Known $\sigma = 1,5$ %. At $\alpha = 0,10$, test if the mean is lower than 6.0 %.

C5

- **Hypotheses:** $H_0: \mu = 6,0$, $H_1: \mu < 6,0$.

- **Test statistic:**

$$z = \frac{5,7 - 6,0}{1,5/\sqrt{100}} = \frac{-0,3}{0,15} = -2,00.$$

- **p-value:** $P(Z \leq -2,00) \approx 0,0228$.
- **Decision:** Reject H_0 since $0,0228 < 0,10$.
- **Conclusion:** Evidence suggests mean inflation expectation is below 6.0 %.

P3 3. Average household electricity bill

A regulator monitors whether the mean monthly electricity bill differs from $\mu_0 = 92$ dollars. Sample: $n = 81$, $\bar{x} = 95,2$, $\sigma = 13,5$. Use $\alpha = 0,05$.

C5

- **Hypotheses:** $H_0: \mu = 92$, $H_1: \mu \neq 92$.

- **Test statistic:**

$$z = \frac{95,2 - 92}{13,5/\sqrt{81}} = \frac{3,2}{1,5} = 2,13.$$

- **p-value:** $2P(Z \geq 2,13) \approx 0,0332$.
- **Decision:** Reject H_0 since $0,0332 < 0,05$.
- **Conclusion:** The mean bill is significantly different from 92 dollars.

P4 4. Mean agricultural loan approval time

A development bank states average approval time is $\mu_0 = 12$ days. For $n = 49$ loans, $\bar{x} = 11,4$ days and known $\sigma = 2,8$ days. At $\alpha = 0,01$, test whether the mean is lower than 12 days.

C5

- **Hypotheses:** $H_0: \mu = 12$, $H_1: \mu < 12$.

- **Test statistic:**

$$z = \frac{11,4 - 12}{2,8/\sqrt{49}} = \frac{-0,6}{0,4} = -1,50.$$

- **p-value:** $P(Z \leq -1,50) \approx 0,0668$.
- **Decision:** Fail to reject H_0 since $0,0668 > 0,01$.
- **Conclusion:** At 1 %, there is insufficient evidence that mean approval time is below 12 days.

P5 5. Average container dwell time at port

Port managers benchmark a mean dwell time of $\mu_0 = 7,5$ days. A sample of $n = 121$ containers has $\bar{x} = 7,1$ days, with known $\sigma = 1,8$ days. At $\alpha = 0,05$, test whether the mean is different from 7.5 days.

C5

- **Hypotheses:** $H_0: \mu = 7,5$, $H_1: \mu \neq 7,5$.

- **Test statistic:**

$$z = \frac{7,1 - 7,5}{1,8/\sqrt{121}} = \frac{-0,4}{0,1636} \approx -2,44.$$

- **p-value:** $2P(Z \leq -2,44) \approx 0,0146$.
- **Decision:** Reject H_0 since $0,0146 < 0,05$.
- **Conclusion:** Mean dwell time differs significantly from 7.5 days.

P6 6. Average daily production of a solar plant

An energy firm claims the plant produces on average $\mu_0 = 420$ MWh per day. Over $n = 36$ days, $\bar{x} = 414$ MWh; known $\sigma = 24$ MWh. At $\alpha = 0,10$, test whether mean production is lower.

C5

- **Hypotheses:** $H_0: \mu = 420$, $H_1: \mu < 420$.

- **Test statistic:**

$$z = \frac{414 - 420}{24/\sqrt{36}} = \frac{-6}{4} = -1,50.$$

- **p-value:** $P(Z \leq -1,50) \approx 0,0668$.
- **Decision:** Reject H_0 since $0,0668 < 0,10$.
- **Conclusion:** Evidence indicates mean production is below 420 MWh.

P7 7. Mean remittance transfer fee

A financial services study tests whether the mean transfer fee is $\mu_0 = 3,2\%$. Sample data: $n = 64$, $\bar{x} = 3,35\%$, known $\sigma = 0,5\%$. At $\alpha = 0,01$, test if the mean is higher.

C5

- **Hypotheses:** $H_0: \mu = 3,2$, $H_1: \mu > 3,2$.

- **Test statistic:**

$$z = \frac{3,35 - 3,2}{0,5/\sqrt{64}} = \frac{0,15}{0,0625} = 2,40.$$

- **p-value:** $P(Z \geq 2,40) \approx 0,0082$.

- **Decision:** Reject H_0 since $0,0082 < 0,01$.

- **Conclusion:** There is strong evidence that mean transfer fees exceed 3.2% .

P8 8. Average freight insurance cost

Analysts test if mean insurance cost per shipment equals $\mu_0 = 1450$ dollars. A sample of $n = 100$ shipments gives $\bar{x} = 1438$, with known $\sigma = 60$. Use $\alpha = 0,05$, two-tailed.

C5

- **Hypotheses:** $H_0: \mu = 1450$, $H_1: \mu \neq 1450$.

- **Test statistic:**

$$z = \frac{1438 - 1450}{60/\sqrt{100}} = \frac{-12}{6} = -2,00.$$

- **p-value:** $2P(Z \leq -2,00) \approx 0,0456$.

- **Decision:** Reject H_0 since $0,0456 < 0,05$.

- **Conclusion:** Mean insurance cost is significantly different from 1450 dollars.

P9 9. Mean wait time for public hospital registration

A ministry target is a mean registration wait of $\mu_0 = 26$ minutes. In $n = 144$ observations, $\bar{x} = 27,1$ minutes and $\sigma = 7,2$ minutes. At $\alpha = 0,10$, test whether mean wait is greater.

C5

- **Hypotheses:** $H_0: \mu = 26$, $H_1: \mu > 26$.

- **Test statistic:**

$$z = \frac{27,1 - 26}{7,2/\sqrt{144}} = \frac{1,1}{0,6} \approx 1,83.$$

- **p-value:** $P(Z \geq 1,83) \approx 0,0336$.

- **Decision:** Reject H_0 since $0,0336 < 0,10$.

- **Conclusion:** Evidence suggests mean registration wait exceeds 26 minutes.

P10 10. Average internet speed in public schools

A broadband program claims mean speed is $\mu_0 = 52$ Mbps. A sample of $n = 49$ schools shows $\bar{x} = 50,8$ Mbps. Assume $\sigma = 6,3$ Mbps. At $\alpha = 0,05$, test whether the mean is lower than 52 Mbps.

C5

- **Hypotheses:** $H_0: \mu = 52$, $H_1: \mu < 52$.

- **Test statistic:**

$$z = \frac{50,8 - 52}{6,3/\sqrt{49}} = \frac{-1,2}{0,9} \approx -1,33.$$

- **p-value:** $P(Z \leq -1,33) \approx 0,0918$.

- **Decision:** Fail to reject H_0 since $0,0918 > 0,05$.

- **Conclusion:** At 5 %, there is insufficient evidence that mean speed is below 52 Mbps.